

Disproof of conjecture 00000000418

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Verdict: FALSE

We disprove conjecture 00000000418 as stated. The conjecture is refuted already at weight $\lambda = (2, 1)$, $n = 3$, and it is refuted under *each* of the two readings of “lattice word” that the conjecture itself supplies. The two readings even disagree with each other: the parenthetical “Yamanouchi (lattice)” makes the two notions synonyms, whereas the second clause defines a lattice word as the reading word of a standard tableau, and reading words are permutations. Since $K_{\lambda, \lambda} = 1$ for every λ , the conjectured value collapses to $f^\lambda/n!$, while the proportion of lattice words among Yamanouchi words is identically 1 under the synonym reading; and under the strict reading a word has to be a permutation, so the only weights with nonzero proportions are the one-column rectangles (1^n) .

1 The conjecture

Quoted verbatim from `conjectures/00000000418.md`:

English. Definition: A Yamanouchi (lattice) word of weight λ is a word in which, in every prefix, the occurrence counts of each letter form a partition; a lattice word is one appearing as the reading word of a standard tableau. Conjecture: The proportion of lattice words of weight λ among Yamanouchi words is exactly $f^\lambda/n! \cdot K_{\lambda, \lambda}^{-1}$, and this ratio attains its maximum $1/2^{n-1}$ when λ is a rectangle.

中文。定义：权 λ 的 *Yamanouchi*(格子) 词指每个前缀中字母 i 的出现次数构成分拆的词；格子词指出现在某标准表读词中的词。猜想：权 λ 的格子词在 *Yamanouchi* 词中的比例恰为 $f^\lambda/n! \cdot K_{\lambda, \lambda}^{-1}$ ，且该比值在 λ 取方框形状时达到最大值 $1/2^{n-1}$ 。

2 Conventions

Definition 1 (word, content, Yamanouchi word). A *word* $w = w_1 w_2 \cdots w_n$ is a finite sequence of positive integers. Its *content* is the vector $\text{content}(w) = (c_1, c_2, \dots)$, where c_i is the number of occurrences of the letter i . A *Yamanouchi word of weight* λ is a word w with $\text{content}(w) = \lambda$ such that for every prefix $w_1 \cdots w_m$ ($0 \leq m \leq n$) the vector of counts $(c_1(m), c_2(m), \dots)$ is a partition, i.e. $c_1(m) \geq c_2(m) \geq \cdots$.

Write $n = |\lambda| = \sum_i \lambda_i$, and write f^λ for the number of standard Young tableaux (SYT) of shape λ and $K_{\lambda, \mu}$ for the Kostka number, i.e. the number of semistandard Young tableaux (SSYT) of shape λ and content μ .

The classical fact on which both readings turn is the standard bijection

$$\{\text{SYT of shape } \lambda\} \longleftrightarrow \{\text{Yamanouchi words of weight } \lambda\},$$

assigning to an SYT T the word $w(T)$ whose i -th letter is the index of the row containing the entry i . For example, for $\lambda = (2, 1)$ the two SYT

$$T_1 = \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & \\ \hline \end{array}, \quad T_2 = \begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & \\ \hline \end{array}$$

give $w(T_1) = 112$ and $w(T_2) = 121$. In particular the number of Yamanouchi words of weight λ is f^λ .

3 Kostka numbers on the diagonal are all 1

Lemma 2. *For every partition λ one has $K_{\lambda,\lambda} = 1$. The unique SSYT of shape λ and content λ is the superstandard tableau, in which every cell of row i contains the entry i .*

Proof. Let T be an SSYT of shape λ and content λ . We show by induction on the row index i that row i of T consists entirely of i 's. For $i = 1$: suppose some cell of a row $r \geq 2$ contained a 1. Row r is weakly increasing, so every cell of row r weakly to the left of that cell also contains a 1; in particular the cell $(r-1, j)$ directly above some cell (r, j) containing 1 exists (since T has shape λ) and satisfies $T(r-1, j) < T(r, j) = 1$ by column strictness, which is impossible because all entries are positive. Hence all 1's lie in the first row. There are exactly λ_1 of them, and the first row has exactly λ_1 cells, so it is filled with 1's. For the induction step, delete the first row; the remaining array is an SSYT of shape $(\lambda_2, \lambda_3, \dots)$ and content $(\lambda_2, \lambda_3, \dots)$ with all entries at least 2; subtracting 1 from every entry and applying the same argument shows that its first row is constant equal to 2, and so on. The superstandard tableau clearly has the required shape, content and semistandardness, so it is the unique one. $\square$

Corollary 3. *The conjectured proportion equals*

$$\frac{f^\lambda}{n! \cdot K_{\lambda,\lambda}} = \frac{f^\lambda}{n!}$$

for every λ .

4 Refutation under reading (A): “Yamanouchi (lattice)” is a synonym

Definition 4 (Reading (A)). The parenthetical in “Yamanouchi (lattice) word” identifies the two terms. Thus a *lattice word of weight λ* means a *Yamanouchi word of weight λ* , and the set of lattice words of weight λ is exactly the set of Yamanouchi words of weight λ .

Theorem 5. *Under reading (A) the proportion of lattice words among Yamanouchi words of weight λ is 1 for every λ , while the conjectured value is $f^\lambda/n!$. At $\lambda = (2, 1)$ one has $1 \neq 1/3$; hence the conjecture is false.*

Proof. Under reading (A) the two classes of words coincide, so the proportion is 1. By Corollary 3 the conjectured value is $f^\lambda/n!$, which is < 1 for every partition of size $n \geq 2$ (see Section 6). Evaluating at $\lambda = (2, 1)$: $n = 3$ and the words of content $(2, 1)$ over $\{1, 2\}$ are 112, 121, 211. The prefix counts are

word	prefix 1	prefix 2	prefix 3	Yamanouchi?
112	(1, 0)	(2, 0)	(2, 1)	yes
121	(1, 0)	(1, 1)	(2, 1)	yes
211	(0, 1)	(1, 1)	(2, 1)	no, since (0, 1) is not a partition

Thus there are exactly 2 Yamanouchi words of weight $(2, 1)$, namely 112 and 121, and the proportion under reading (A) is

$$\frac{2}{2} = 1.$$

The hook lengths of the cells of $(2, 1)$ are 3, 1, 1, so the hook-length formula gives

$$f^{(2,1)} = \frac{3!}{3 \cdot 1 \cdot 1} = \frac{6}{3} = 2,$$

in agreement with the two SYT T_1, T_2 above; moreover $n! = 6$ and $K_{(2,1),(2,1)} = 1$ by Lemma 2. The conjectured value is therefore

$$\frac{f^{(2,1)}}{3! \cdot K_{(2,1),(2,1)}} = \frac{2}{6 \cdot 1} = \frac{1}{3}.$$

Since

$$1 \neq \frac{1}{3},$$

and cross-multiplying gives

$$2 \cdot 6 = 12 \neq 4 = 2 \cdot 2,$$

the conjecture fails at $\lambda = (2, 1)$ under reading (A). □

The enumeration is reproduced by brute force in `reproduce.py` and is also formalised in `lean4/` (the Yamanouchi words of weight $(2, 1)$ are proved to be exactly 112 and 121, and the two readings are proved to give the proportions $2/2$ and $0/2$).

4.1 The asserted maximum $1/2^{n-1}$

Reading (A) also kills the second clause. At $\lambda = (2, 1)$ one has $n = 3$, so the asserted maximum is $1/2^{n-1} = 1/4$. But the proportion under reading (A) is 1, and

$$1 > \frac{1}{4},$$

so the asserted maximum is not a maximum at all: it is exceeded. Equivalently, the conjectured value $f^\lambda/n!$ would have to attain $1/2^{n-1}$ at a rectangle, but the proportion itself is 1 for every shape, and $1 \neq 1/4$.

5 Refutation under reading (B): reading words of standard tableaux

Definition 6 (Reading (B)). Read the second clause literally: a *lattice word* is a word that appears as the reading word of a standard tableau, i.e. the word obtained by reading the entries of some SYT row by row (say from bottom to top, each row left to right).

Theorem 7. *Under reading (B) there is no lattice word of weight $(2, 1)$, so the proportion of lattice words among the 2 Yamanouchi words of weight $(2, 1)$ is*

$$\frac{0}{2} = 0 \neq \frac{1}{3}.$$

Hence the conjecture is false.

Proof. In a standard tableau the entries $1, 2, \dots, n$ each occur exactly once. Hence every reading word is a permutation of $\{1, \dots, n\}$ and has content $(1, 1, \dots, 1) = (1^n)$, so it can have weight $(2, 1)$ only if $(2, 1) = (1, 1, 1)$, which is false. Concretely, the two SYT of shape $(2, 1)$ have reading words 312 and 213 (bottom row first), both of content $(1, 1, 1)$. Therefore the number of lattice words of weight $(2, 1)$ under reading (B) is 0, and the proportion is $0/2 = 0$, whereas the conjectured value is $1/3$ by Theorem 5. This proves the claim. $\square$

Remark 8. The two readings are mutually inconsistent already at the level of the definition: Yamanouchi words of weight λ may repeat letters (whenever some $\lambda_i \geq 2$), while reading words of standard tableaux never do. At $\lambda = (2, 1)$ both readings are fatal to the conjecture, and they fail in opposite directions (1 versus 0, against the claimed $1/3$).

6 The general failure: $K_{\lambda, \lambda} = 1$ makes the value $f^\lambda/n!$

By Lemma 2, $K_{\lambda, \lambda} = 1$ for every partition, so the conjectured value is always $f^\lambda/n!$. Under reading (A) the true proportion is 1 for every λ . Since

$$\sum_{\lambda \vdash n} f^\lambda = n! \quad \text{and} \quad f^\lambda \geq 1 \text{ for every } \lambda \vdash n,$$

for $n \geq 2$ there are at least two partitions of n , hence $f^\lambda < n!$ and therefore $f^\lambda/n! < 1$. So the conjectured proportion $f^\lambda/n!$ differs from the true proportion 1 for *every* partition λ of every size $n \geq 2$; the only case that is not immediately excluded is $n = 1$ ($\lambda = (1)$, $f^\lambda/n! = 1 = 1$), which is trivial.

6.1 The maximisers of $f^\lambda/n!$ are not rectangles

The value $f^\lambda/n!$ is the probability that a uniformly random permutation of $\{1, \dots, n\}$ has RSK shape λ . Its maximisers for small n are:

n	maximising λ	f^λ	$f^\lambda/n!$
3	(2, 1)	2	$2/6 = 1/3$
5	(3, 1, 1)	6	$6/120 = 1/20$
6	(3, 2, 1)	16	$16/720 = 1/45$

None of these is a rectangle. At $n = 3$ the two rectangle partitions are (3) and (1, 1, 1), both with $f^\lambda = 1$ and $f^\lambda/n! = 1/6 < 1/3$; the maximum is attained at the non-rectangle (2, 1). So the

clause “attains its maximum $1/2^{n-1}$ when λ is a rectangle” fails in two independent ways: the true proportion under reading (A) is $1 > 1/2^{n-1}$ for every $n \geq 2$, and the conjectured value $f^\lambda/n!$ is not maximised at a rectangle and does not equal $1/2^{n-1}$ in general (at $n = 3$ its maximum is $1/3 \neq 1/4$). The case $n = 2$ is the only coincidence ($f^{(2)} = f^{(1,1)} = 1$, $n! = 2$, and $1/2! = 1/2 = 1/2^{n-1}$), but there the true proportion under reading (A) is still $1 \neq 1/2$, so the conjecture fails there as well.

7 Reproduction and formalisation

The numerical checks are performed by `reproduce.py` (Python 3, standard library only), which enumerates all words of content $(2, 1)$, classifies them by the prefix condition, lists the reading words of the two SYT of shape $(2, 1)$, computes $f^{(2,1)}$ by the hook-length formula, computes $K_{(2,1),(2,1)}$ by brute-force enumeration of semistandard tableaux, asserts both proportions differ from $1/3$, extends the computation to all partitions of size $n \leq 6$ (verifying $\#Yamanouchi = f^\lambda = \#SYT$ and $K_{\lambda,\lambda} = 1$ throughout), and prints `PASS` with exit status 0.

The refutation is formalised in Lean 4 in the accompanying `lean4/` project (core Lean only, `import Std`, no `Mathlib`, no `sorry`, toolchain `leanprover/lean4:v4.33.1`). The formal statements include

```
theorem yamWords21_eq : yamWords21 = [[1,1,2],[1,2,1]]
theorem card_yamWords21 : yamWords21.length = 2
theorem yam_eq_latticeReadings : yamWords21 = latticeReadings21
theorem numLattice21_B_eq : numLattice21_B = 0
theorem f21_eq : f21 = 2
theorem K21_eq : K21 = 1
theorem proportionA_ne_claim : propNumA * claimDen ≠ claimNum * propDenA
theorem cross_values : propNumA * claimDen = 12 ∧ claimNum * propDenA = 4
theorem proportionB_ne_claim : propNumB * claimDen ≠ claimNum * propDenB
theorem quarter_lt_one : (1 : Nat) * 1 < 4 * 1
theorem proportionA_exceeds_max : propNumA * 4 > 1 * propDenA
theorem conjecture_00000000418_refuted : ...
```

Here `proportionA_ne_claim` is the cross-multiplied form of $2/2 \neq 2/6$ (namely $12 \neq 4$) and `proportionA_exceeds_max` is the cross-multiplied form of $1 > 1/4$. The document is built with `tectonic main.tex`; the resulting PDF is `build/main.pdf`.